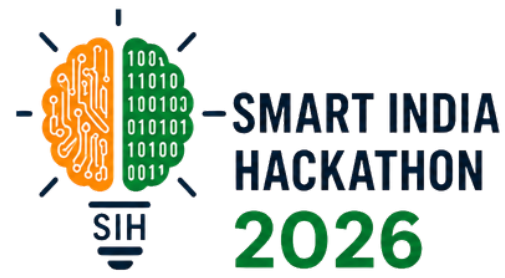
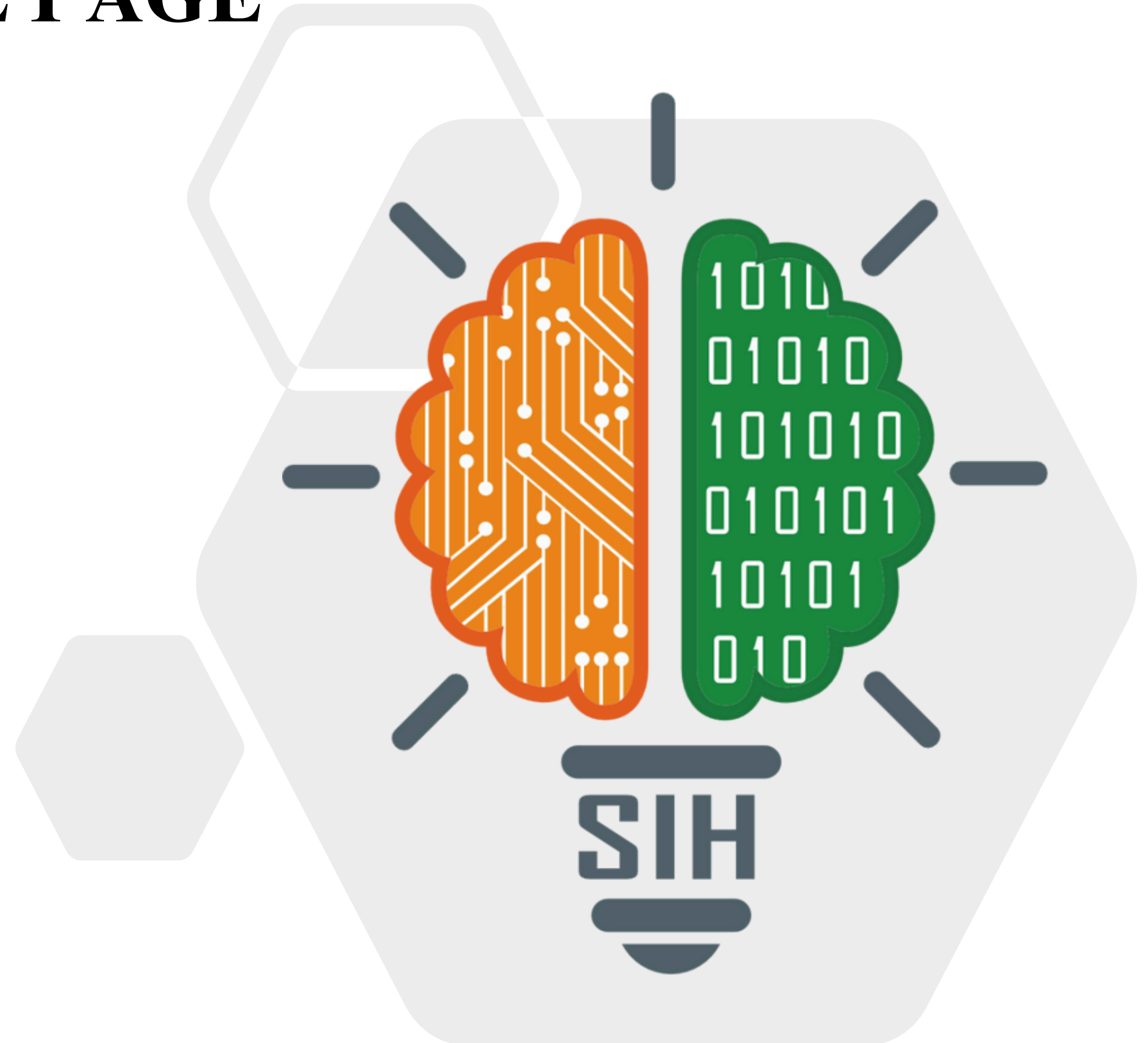


SMART INDIA HACKATHON 2026



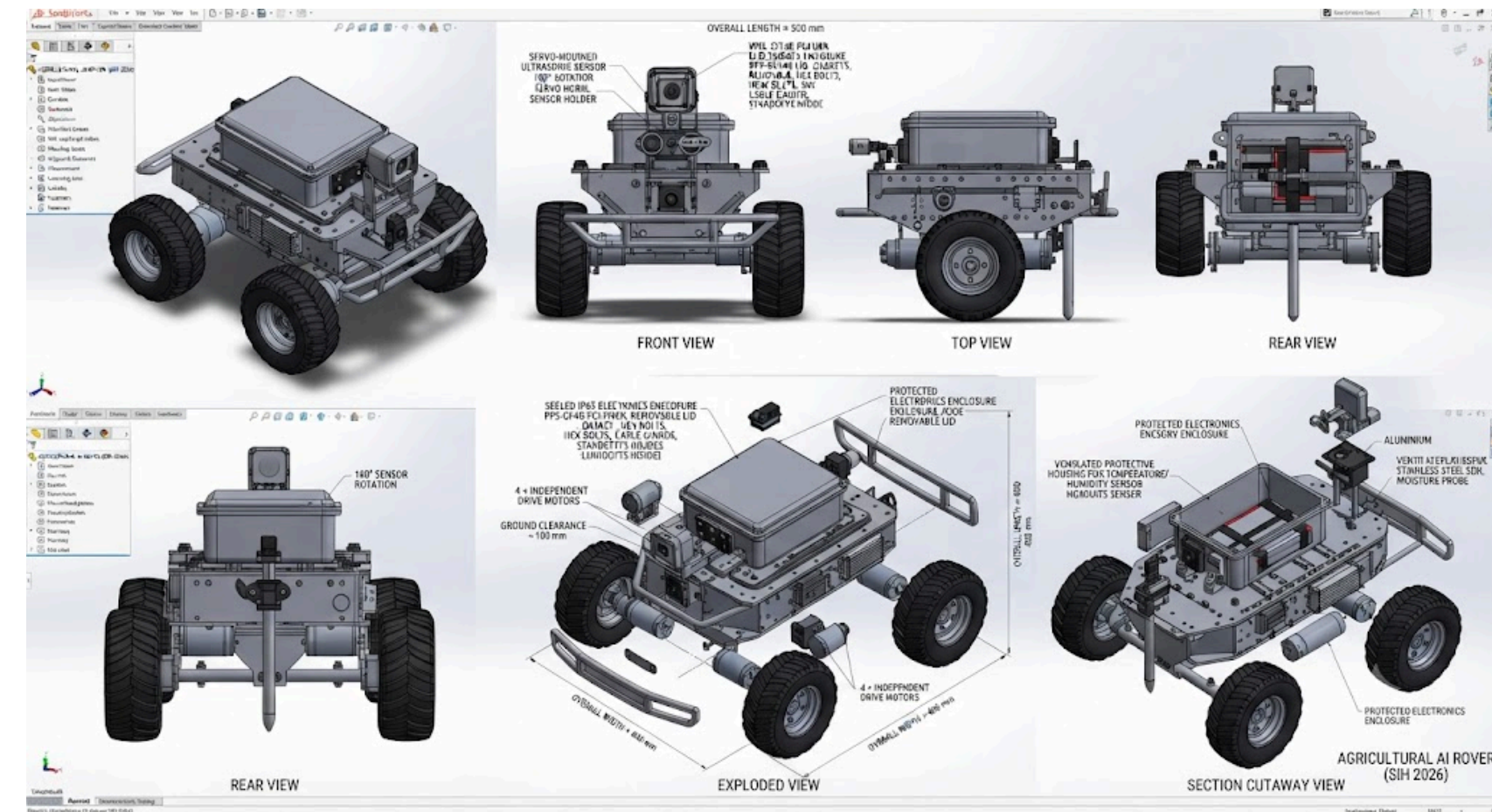
TITLE PAGE

- **Problem Statement ID – SIH26039**
- **Problem Statement Title - TARA – Terrain Analysis & Rescue Assistant**
- **Theme- Tourism / Mine Safety**
- **PS Category- Hardware**
- **Team ID- 338**
- **Team Name : C-66**



“Detect Early. Respond Fast. Protect Lives.”

- **Rugged Rover for Hazardous Mines — Remote inspection where humans face risk.**
- **Sense Every Threat — Gas, CO, temperature, humidity, vibration & obstructions.**
- **Detect & React Instantly — Smart thresholds trigger alarms and automated ventilation.**
- **Stay Connected Underground — LoRa transmits critical data to the surface.**
- **Alert. Monitor. Protect. — Live status + GSM SMS alerts enable rapid operator response.**



Components Used

Sensor Array Integration

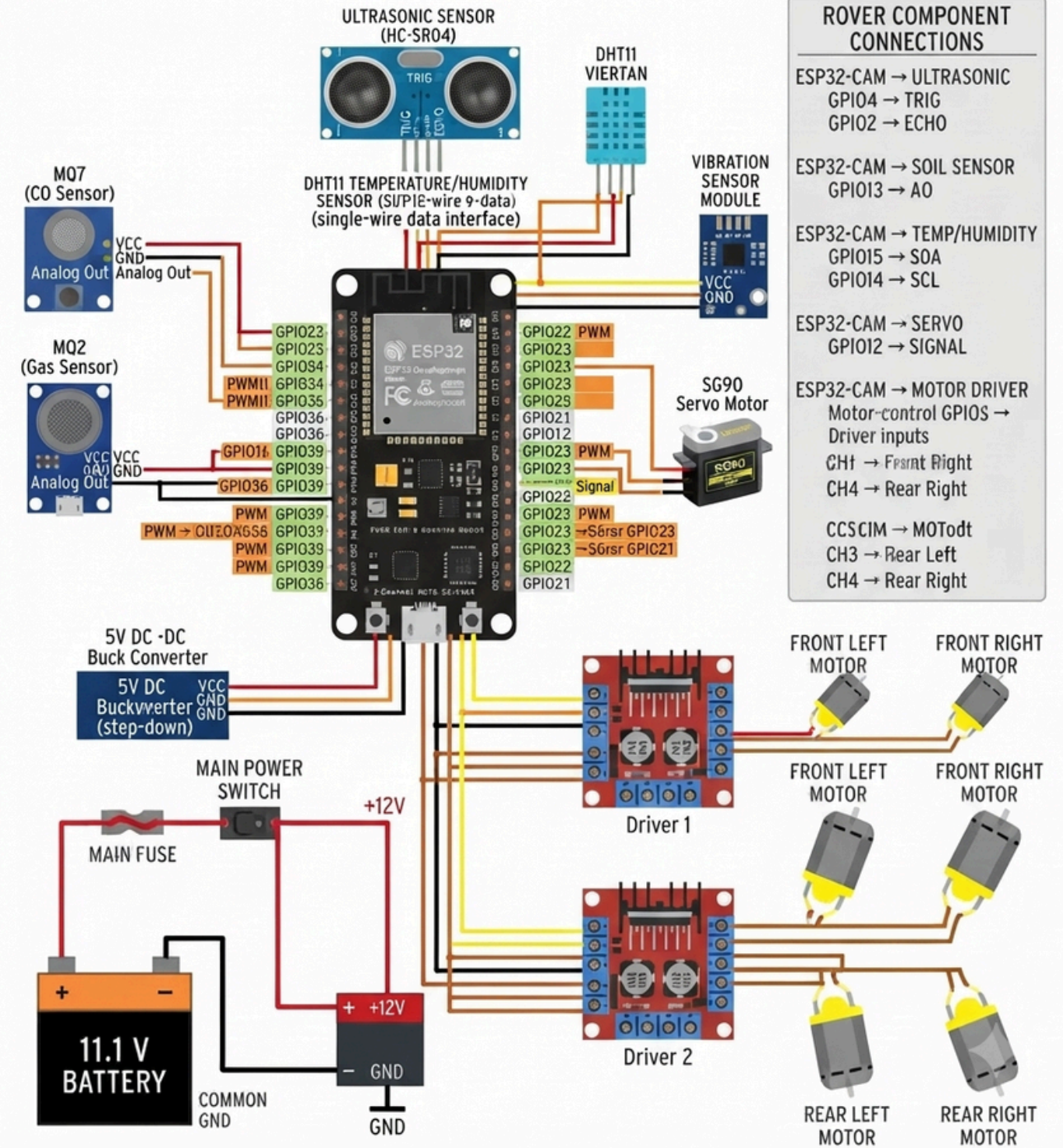
- MQ-2: Combustible gas detection
- ↓
- MQ-7: CO detection
- ↓
- DHT22: Temperature & humidity
- ↓
- HC12: Structural vibration
- ↓
- HC-SR04: Obstruction/clearance

Communication and Control

- LoRa SX1278: Underground communication
- ↓
- ESP32: Surface relay/controller
- ↓
- SIM800L: SMS alerts

System Output and Power

- OLED SSD1306: Status display
- ↓
- Relay + 5V Fan: Automated ventilation
- ↓
- 18650 + TP4056: Power system



KEY FEASIBILITY ADVANTAGES

LOW COST AND AVAILABLE
COMPONENTS

MULTI-SENSOR REAL-TIME
HAZARD DETECTION

LoRa-BASED UNDERGROUND
COMMUNICATION

MODULAR DESIGN FOR ROVER
INTEGRATION

SCALABLE TO MULTIPLE MINE
ZONES

POTENTIAL CHALLENGES AND RISKS

ROUGH , CONFINED
UNDERGROUND TERRAIN

DUST , MOISTURE AND VIBRATION
IMPACT

OVERCOMING STRATEGIES

USE RUGGED , MINE-GRADE
HARDWARE AND ENCLOSURE

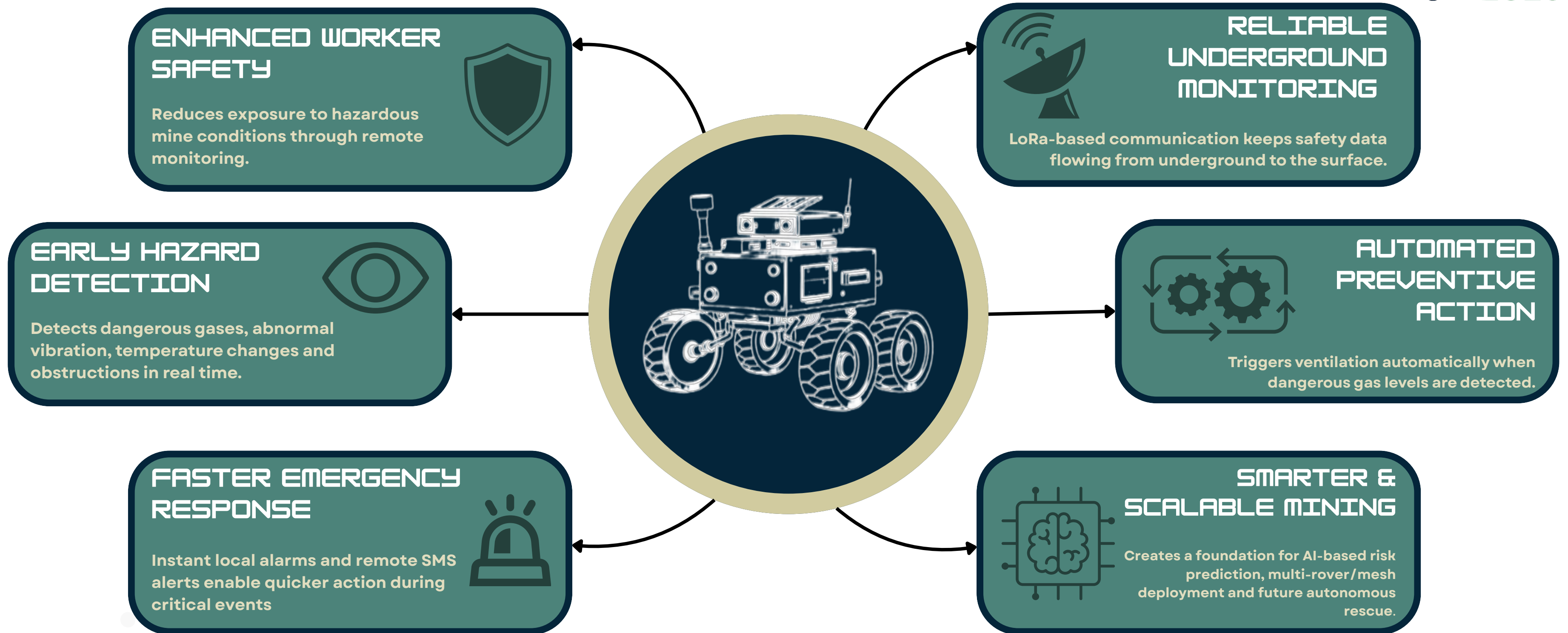
PAIR UP LoRa WITH SURFACE
RELAY FOR RELIABLE
COMMUNICATION

OPTIMIZE POWER USING LOW-
DUTY-CYCLE SENSING

CALIBRATE SENSORS AND USE
ADAPTIVE THRESHOLDS

SCALE THROUGH LoRa MESH
AND AI BASED ANOMALY
DETECTION

IMPACT AND BENEFITS



RESEARCH AND REFERENCES

- 1) <https://ieeexplore.ieee.org/abstract/document/11012562/>
- 2) <https://link.springer.com/article/10.1007/s00542-023-05484-z>
- 3) https://www.researchgate.net/publication/404370723_An_ESP32-BasedAutonomous_Mobile_Robot_for_RoughTerrain_Navigation_Using_Multi-Sensor_Obstacle_Avoidance_and_Hazard_Detection